

Comprehensive Search Results (2022–2025)

Below is a CSV-formatted list of peer-reviewed papers (2022–2025) on **“Automated Compliance Checking” AND Ontology** across major publishers. Each entry includes Title, Authors, Year, DOI, Abstract, and Source (journal or conference name):

Title	Authors	Year	DOI	Abstract	Source
Knowledge-informed semantic alignment and rule interpretation for automated compliance checking	Zhe Zheng; Yu-Cheng Zhou; Xin-Zheng Lu; Jia-Rui Lin,	2022	10.1016/j.autcon.2022.104524	"As an essential procedure to improve design quality in the construction industry, automated rule checking (ARC) requires intelligent rule interpretation from regulatory texts and precise alignment of concepts from different sources. However, there still exists semantic gaps between design models and regulatory texts, hindering the exploitation of ARC. Thus, a knowledge-informed framework for improved ARC is proposed based on natural language processing. Within the framework, an ontology is first established to represent domain knowledge, including concepts, synonyms, relationships, constraints, etc. Then, semantic alignment and conflict resolution are introduced to enhance the rule interpretation process based on predefined domain knowledge and unsupervised learning techniques ¹ ² ."	Automation in Construction
Automated code compliance checking research based on BIM and knowledge graph	Junlong Peng; Xiangjun Liu,	2023	10.1038/s41598-023-34342-1	"Automated code compliance checking plays an important role in moving the construction industry forward. While traditional drawing review relies on the identification of industry experts, the purpose of this study is to realize automatic code review by using BIM technology and knowledge graph technology. The method is based on knowledge graph to transform the specification provisions involved in the review of drawings into a computer-recognizable structured language using natural language processing technology to form a human-machine-readable knowledge graph pattern. For the review of BIM models, the BIM model information is exported and the global resource description framework of the building model is obtained in Protégé, and the review report is finally obtained after mapping rules and review rules. Finally, the feasibility of the method is verified by an example ³ ⁴ ."	Scientific Reports
Rule capture of automated compliance checking of building requirements: a review	Zijing Zhang; Ling Ma; Tim Broyd,	2023	10.1680/jsmic.23.00005	"In the architectural, engineering and construction industry, building design needs to be checked against regulations before it can be finalised and progress to the construction stage. The traditional manual compliance-checking process is error prone and time consuming. As a solution, automated compliance checking (ACC) was proposed. Rule capture is a crucial bottleneck of ACC. Despite many studies in this domain, no research has synthesised the themes and identified future research opportunities. This paper aims to fill this gap by conducting a systematic literature review and identifying challenges facing this field. The findings revealed that the rule capture process had attracted interest in	

the past years, and more semi-automated and automated methods have been proposed. The current representation development process lacks a methodological backdrop... Further research is required to address these issues ⁵ ⁶.", Proceedings of the ICE - Smart Infrastructure and Construction Compliance Checking on Topological Spatial Relationships of Building Elements Based on Building Information Models and Ontology, Xuefeng Zhao; Lingli Huang; Zhe Sun; Xiongtao Fan; Meng Zhang, 2023, 10.3390/su151410901, "Compliance checking on the topological spatial relationships of building elements is vital for ensuring the safety and the quality of buildings. However, the complex topological spatial relationships of buildings are not usually expressed in the design scheme directly. Manual checking is still needed to analyze the design scheme and extract the spatial relationships. Such manual checking is always time consuming and prone to error. Therefore, this study has proposed a compliance checking method based on a building information model (BIM) and building ontologies for the automatic checking of topological spatial relationships. Firstly, the topological spatial relationships are well captured and represented according to the location relation of building elements... The authors have validated the proposed method through a case study. The results show that the proposed method could help to achieve automatic compliance checking on topological spatial relationships of building elements ⁷ ⁸.", Sustainability

NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR, Orlando Amaral Cejas; Muhammad Ilyas Azeem; Sallam Abualhaija; Lionel Briand, 2023, 10.1109/TSE.2023.3288901, "When the entity processing personal data (the processor) differs from the one collecting personal data (the controller), processing personal data is regulated in Europe by the GDPR through data processing agreements (DPAs). Checking the compliance of DPAs contributes to the compliance verification of software systems as DPAs are an important source of requirements... However, manually checking whether a given DPA complies with GDPR is challenging as it requires significant time and effort. In this paper, we propose an automated solution to check the compliance of a given DPA against GDPR... our approach automatically generates phrasal-level representations for the textual content of the DPA and compares them against predefined representations of the "shall" requirements... the approach not only assesses whether the DPA is GDPR compliant but it further provides recommendations about missing information in the DPA ⁹ ¹⁰.", IEEE Transactions on Software Engineering

Optimizing automated compliance checking with ontology-enhanced natural language processing: Case in the fire safety domain, Yian Chen; Huixian Jiang, 2024, 10.1016/j.jenvman.2024.123320, "The fire safety compliance checking (FSCC) plays a crucial role in ensuring the quality of fire engineering design and eliminating inherent fire hazards. It requires an objective and rational interpretation of fire regulations. However, the texts of fire regulations are filled with numerous rules related to spatial limitations, which pose a significant challenge in interpreting them. The current method of interpreting these rules mostly relies on manual translation, which is not efficient. To address this issue, this study proposes an innovative automated framework for interpreting rules by combining ontology technology with natural language processing (NLP)... The tree structure is then mapped to logical clauses based on semantic consistency, thereby improving the efficiency of interpretation. Experimental results demonstrate that the

architecture achieves an F1 score of 86.27 for entity extraction and 81.81 for spatial relationship extraction tasks, with an accuracy of 96.26% in the formalization of logical rules ¹¹ ¹² .", *Journal of Environmental Management*

Semantic web-based automated compliance checking with integration of Finite Element analysis, Panagiotis Patlakas; Ioannis Christovasilis; Lorenzo Riparbelli; Franco K. T. Cheung; Edlira Vakaj, 2024, 10.1016/j.aei.2024.102448, "Automatic Compliance Checking (ACC) is a promising response to the challenges involved in meeting building and planning regulations, and increasingly utilised by researchers in the context of Building Information Modelling (BIM) and the Industry Foundation Classes (IFC). However, engineers often use computational methods, such as Finite Element Analysis (FEA), to solve challenging engineering problems. Such methods have received little attention in the context of ACC and BIM. The work presented here proposes a holistic approach, integrating FEA in the IFC schema, and utilising a Semantic Web-based system for ACC. A top-level ontology, the Building Compliance Ontology (BCO), is proposed, with the aim of covering all aspects of compliance in the Built Environment... Compliance is checked automatically by running SHACL scripts in a Semantic Web environment ¹³ ¹⁴ .", *Advanced Engineering Informatics*

Automated Building Information Modeling Compliance Check through a Large Language Model Combined with Deep Learning and Ontology, Nanjiang Chen; Xuhui Lin; Hai Jiang; Yi An, 2024, 10.3390/buildings14071983, "Ensuring compliance with complex industry standards and regulations during the design and implementation phases of construction projects is a significant challenge in the building information modeling (BIM) domain. Traditional manual compliance checking methods are inefficient and error-prone, failing to meet modern engineering demands. Natural language processing (NLP) and deep learning methods have improved efficiency and accuracy in rule interpretation and compliance checking. However, these methods still require extensive manual feature engineering, large annotated datasets, and significant computational resources. Large language models (LLMs) provide robust language understanding with minimal labeled data due to their pre-training and few-shot learning capabilities... This study introduces an innovative automated compliance checking framework that integrates LLM, deep learning models, and ontology knowledge models... This novel combination of deep learning and LLM significantly enhances the efficiency and accuracy of compliance checks by automating the processing of regulatory texts and reducing manual intervention ¹⁵ ¹⁶ .", *Buildings*

Automated Compliance Checking System for Structural Design Codes in a BIM Environment, Wonhui Goh; Jaeguk Jang; Sang I. Park; Bong-Hyuck Choi; Heuiseok Lim; Goangseup Zi, 2024, 10.1007/s12205-024-1121-5, "This study proposes a framework for developing an automated compliance checking (ACC) system that supports the verification of structural calculations in openBIM format. The framework consists of four main aspects: (1) classification of design codes depending on structural components and limit states to be checked; (2) preparation of a structural model according to Industry Foundation Classes; (3) interpretation of conditions in design codes into a machine-readable rule language; and (4) design of an ACC system, with a focus on the rule flow in the structural design code. The framework was demonstrated through a detailed example of an ACC system using ifcOWL, SWRL, and the Drools inference engine... The proposed framework can help develop ACC systems in structural

engineering and can be customized to meet user requirements and accommodate various data sources and design codes ¹⁷ ¹⁸ .",KSCE Journal of Civil Engineering

A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data,Joseph Anim; Livio Robaldo; Adam Z. Wyner,2024,10.3390/

info15120759,"Automated Compliance Checking (ACC) has emerged as a critical tool for enforcing legal regulations across various domains. This paper contributes to ongoing research in Semantic Web technologies, particularly focusing on the execution of SHACL-SPARQL rules on RDF data. The RDF, being one of the most widely used knowledge representation formats, serves as the foundation of our approach, ensuring compatibility with existing standards and enhancing interoperability. Our research enhances the aggregate and temporal aspects of ACC by addressing the limitations of traditional ACC methodologies, which often fall short in managing the nuanced temporal and aggregate requirements essential for legal reasoning. Through a case study analysis of selected regulations with aggregate and temporal facets... the findings underscore the potential of Semantic Web technologies to transform ACC practices by moving towards standardized, interoperable solutions ¹⁹

²⁰ .",Information

Semantic and ontology-based analysis of regulatory documents for construction industry digitalization,Zarina Kabzhan; Alexandr Shakhnovich; Sergey Gorshkov; Yussuf Yemenov; Fedor Gorshkov; Nazym Shogelova,2025,10.3389/fbuil.2025.1575913,"In the context of the digitalization of the construction industry, the demand for automation in the analysis of regulatory documents is increasing. The significant volume, structural complexity, and frequent amendments of regulatory acts lead to semantic inconsistencies, duplication of provisions, and contradictions in requirements. The aim of this study is to develop a method for the automated analysis of regulatory documents in the construction sector based on the integration of ontological modeling and natural language processing (NLP) techniques. The study proposes a methodology for constructing semantic "profiles" of regulatory statements, which include structured components: subject, predicate, object, modality, and additional conditions. A software prototype has been developed, implementing an algorithm for semantic matching of regulatory requirements using an ontological model that incorporates SKOS-based terminology, deontic logic, and domain-specific concepts of the construction sector. Experiments were conducted on a corpus of 14 regulatory documents... demonstrating high computational efficiency and acceptable quality metrics (F1-score up to 0.86). The results confirm the applicability of the proposed method for integration into automated regulatory compliance control systems ²¹ ²² .",Frontiers in Built Environment

OntoBPR: An ontology-based framework for performing building permit reviews using standardized information containers,Philipp Hagedorn; Judith Fauth; Sven Zentgraf; Ioannis Brilakis; Sebastian Seiss; Markus König,2025,10.1016/j.aei.2025.103369,"Building permitting is essential for ensuring the safety, sustainability, and societal alignment of construction projects. Despite interest from both practitioners and researchers, the process remains largely manual and fragmented. Ontologies offer a promising solution by managing complexity and enabling automation through semantic information, though current ontologies in the building permit domain are limited to specific aspects like building code checking. On the process level, the OntoBPR

framework integrates multiple domain-specific ontologies for a seamless digital permitting process and provides a workflow to automate the lifecycle of the permit review. Therefore, it suggests integrating the submitted building application using standardized information containers. The paper explores how digital applications can be submitted, reviewed, verified for completeness, and forwarded to authorities, and how permit review results can be gathered to support decision-making and automate notification issuance... OntoBPR formalizes a multi-layered ontology that advances and aligns the partitioned building permit process and provides an adaptable framework to harmonize diverse legal, informatics, and procedural aspects ²³ ²⁴ .", Advanced Engineering Informatics

Each entry above is drawn from peer-reviewed journals or conferences (IEEE, ACM/Elsevier, Springer, MDPI, Frontiers, Nature, ICE etc.), and the abstracts are **properly cited** to the source literature as evidence.

¹ ² Knowledge-Informed Semantic Alignment and Rule Interpretation for Automated Compliance Checking - Jia-Rui Lin's Page

<https://linjiarui.net/en/publications/2022-08-03-knowledge-informed-semantic-alignment-and-rule-interpretation-for-ACC>

³ ⁴ Automated code compliance checking research based on BIM and knowledge graph | Scientific Reports

https://www.nature.com/articles/s41598-023-34342-1?error=cookies_not_supported&code=12896577-f1cd-4af3-9fb0-937a91b4c5a3

⁵ ⁶ Rule capture of automated compliance checking of building requirements: a review - UCL Discovery

<https://discovery.ucl.ac.uk/id/eprint/10174755/>

⁷ ⁸ Compliance Checking on Topological Spatial Relationships of Building Elements Based on Building Information Models and Ontology

<https://www.mdpi.com/2071-1050/15/14/10901>

⁹ ¹⁰ ORBilu: Detailed Reference

<https://orbilu.uni.lu/handle/10993/55409?&locale=en>

¹¹ ¹² Optimizing automated compliance checking with ontology-enhanced natural language processing: Case in the fire safety domain - PubMed

<https://pubmed.ncbi.nlm.nih.gov/39547025/>

¹³ ¹⁴ Semantic web-based automated compliance checking with integration of Finite Element analysis

<https://flore.unifi.it/handle/2158/1407193>

¹⁵ ¹⁶ Automated Building Information Modeling Compliance Check through a Large Language Model Combined with Deep Learning and Ontology

<https://www.mdpi.com/2075-5309/14/7/1983>

¹⁷ ¹⁸ Automated Compliance Checking System for Structural Design Codes in a BIM Environment | KSCE Journal of Civil Engineering

https://link.springer.com/article/10.1007/s12205-024-1121-5?error=cookies_not_supported&code=7749053c-54e9-47c1-8acc-fd0a996b744c

¹⁹ ²⁰ A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data

<https://www.mdpi.com/2078-2489/15/12/759>

²¹ ²² Frontiers | Semantic and ontology-based analysis of regulatory documents for construction industry digitalization

<https://www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2025.1575913/full>

²³ ²⁴ OntoBPR: An ontology-based framework for performing building permit reviews using standardized information containers

<https://www.repository.cam.ac.uk/items/688b8820-9a14-46f5-b32f-589b5f0a18c5>